

Literature-implied answer for arXiv:1703.02919

Exact simultaneous bisection follows from Lyapunov convexity

Status. `literature_implied_answer` (full affirmative answer to Remark 3.4). This is not a new theorem produced by the run. It is a direct specialization of the classical finite-dimensional Lyapunov convexity theorem.

Source question. In Rainis Haller, Paavo Kuuseok, and Märt Põldvere, *On convex combinations of slices of the unit ball in Banach spaces*, arXiv:1703.02919v2, Remark 3.4 on PDF page 12 follows Lemma 3.2. Given atomless finite nonnegative measures $\mu_1, \dots, \mu_n$ on a σ -algebra and E measurable, the lemma finds a partition $E = A \dot{\cup} B$ satisfying

$$|\mu_i(A) - \mu_i(B)| \leq \varepsilon \quad (1 \leq i \leq n).$$

The remark asks whether one may take $\varepsilon = 0$ when $n \geq 2$.

Supporting literature. Lyapunov's convexity theorem says that the range of a finite vector measure in a Euclidean space is compact and is convex if the vector measure is atomless. This formulation appears explicitly in the abstract and introduction on PDF page 1 of Peng Dai and Eugene A. Feinberg, *Extension of Lyapunov's Convexity Theorem to Subranges*, arXiv:1102.2534. Their introduction defines a vector measure $(\mu_1, \dots, \mu_m)$ to be atomless when every coordinate measure is atomless, exactly matching the hypotheses of Lemma 3.2.

Identification. Restrict the measures to E and define the finite-dimensional vector measure

$$\nu(C) := (\mu_1(C), \dots, \mu_n(C)), \quad C \subseteq E, \ C \text{ measurable.}$$

Every coordinate is atomless, so Lyapunov's theorem makes the range

$$R_\nu(E) := \{\nu(C) : C \subseteq E, \ C \text{ measurable}\}$$

convex. It contains both $0 = \nu(\emptyset)$ and $\nu(E)$. Hence it contains their midpoint $\frac{1}{2}\nu(E)$. Choose measurable $A \subseteq E$ with $\nu(A) = \frac{1}{2}\nu(E)$ and put $B = E \setminus A$. Then, for every i ,

$$\mu_i(A) = \frac{1}{2}\mu_i(E) = \mu_i(B).$$

Answer. *Yes. For every finite family of atomless finite nonnegative measures and every measurable E , there is an exact simultaneous bisection $E = A \dot{\cup} B$ with $\mu_i(A) = \mu_i(B)$ for all i .*

Provenance and scope. The supporting statement predates the source question by six years. Bounded search of the run indexes, exact source phrase, paper title together with “Lyapunov”, and lemma/author combinations found no later paper explicitly identifying Remark 3.4. The answer is therefore recorded as literature-implied rather than as a new proof or an explicitly announced later answer. This packet concerns only the measure-theoretic remark.

References

- [1] R. Haller, P. Kuuseok, and M. Põldvere, *On convex combinations of slices of the unit ball in Banach spaces*, arXiv:1703.02919v2 (2017).
- [2] P. Dai and E. A. Feinberg, *Extension of Lyapunov's Convexity Theorem to Subranges*, arXiv:1102.2534 (2011).
- [3] A. A. Lyapunov, *Sur les fonctions-vecteurs complètement additives*, Izv. Akad. Nauk SSSR Ser. Mat. 4 (1940), 465–478.